

Topic 15: Organic Chemistry: Carbonyls, Carboxylic Acids and Chirality

15A: Chirality

Students will be assessed on their ability to:

15.1	know that optical isomerism is a result of chirality in molecules with a single chiral centre
15.2	understand that optical isomerism results from chiral centre(s) in a molecule with asymmetric carbon atom(s) and that optical isomers (enantiomers) are object and non-superimposable mirror images and be able to draw 3D diagrams of these optical isomers
15.3	know that optical activity is the ability of a single optical isomer to rotate the plane of polarisation of plane-polarised monochromatic light in molecules containing a single chiral centre
15.4	know what is meant by the term 'racemic mixture'
15.5	be able to use data on optical activity of reactants and products as evidence for S _N 1 and S _N 2 mechanisms and addition to carbonyl compounds

15B: Carbonyl compounds

Students will be assessed on their ability to:

15.6	understand the nomenclature of aldehydes and ketones and be able to draw their structural, displayed and skeletal formulae
15.7	understand that aldehydes and ketones: i do not form intermolecular hydrogen bonds and this affects their physical properties ii can form hydrogen bonds with water and this affects their solubility
15.8	understand the reactions of carbonyl compounds with: i Fehling's or Benedict's solution, Tollens' reagent and acidified dichromate(VI) ions <i>In equations, the oxidising agent can be represented as [O].</i> ii lithium tetrahydridoaluminate(III) (lithium aluminium hydride) in dry ether (ethoxyethane) <i>In equations, the reducing agent can be represented by [H].</i> iii HCN, in the presence of KCN, as a nucleophilic addition reaction, using curly arrows, relevant lone pairs, dipoles and evidence of optical activity to show the mechanism iv 2,4-dinitrophenylhydrazine (2,4-DNPH), as a qualitative test for the presence of a carbonyl group and to identify a carbonyl compound given data of the melting temperatures of derivatives <i>The equation for this reaction is not required.</i> v iodine in the presence of alkali (the iodoform test)
	Further suggested practical: Reactions of aldehydes and ketones given in 15.8 i, iv and v

15C: Carboxylic acids

Students will be assessed on their ability to:

15.9	understand the nomenclature of carboxylic acids and be able to draw their structural, displayed and skeletal formulae
15.10	understand that hydrogen bonding affects the physical properties of carboxylic acids, in relation to their boiling temperatures and solubility
15.11	understand that carboxylic acids can be prepared by the oxidation of alcohols or aldehydes and the hydrolysis of nitriles
15.12	understand the reactions of carboxylic acids with: i lithium tetrahydridoaluminate(III) (lithium aluminium hydride) in dry ether (ethoxyethane) ii bases to produce salts iii phosphorus(V) chloride (phosphorus pentachloride) iv alcohols in the presence of an acid catalyst
	Further suggested practicals: i solubility of a range of carboxylic acids, aldehydes and ketones ii preparation of carboxylic acids by the oxidation of alcohols and aldehydes iii reactions of carboxylic acids given in 15.12 ii, iii and iv

15D: Carboxylic acid derivatives

Students will be assessed on their ability to:

15.13	understand the nomenclature of acyl chlorides and esters and be able to draw their structural, displayed and skeletal formulae
15.14	understand the reactions of acyl chlorides with: i water ii alcohols iii concentrated ammonia iv amines
15.15	understand the hydrolysis reactions of esters, in acidic and alkaline solution
15.16	understand how polyesters, such as terylene, are formed by condensation polymerisation reactions.
	Further suggested practicals: i demonstration of the reactions of ethanoyl chloride given in 15.14 i, ii and iii ii the preparation of esters such as ethyl ethanoate as a solvent or a pineapple flavouring iii hydrolysis of an ester

15E: Spectroscopy and chromatography

Knowledge of the concepts introduced in Unit 2, Topic 10D: Mass spectra and IR will be assumed and extended in this topic

Students will be assessed on their ability to:

15.17	be able to use data from mass spectra to: - i suggest possible structures of a simple organic compound given accurate relative molecular masses - ii calculate the accurate relative molecular mass of a compound, given accurate relative atomic masses to four decimal places
15.18	understand that carbon-13, (^{13}C) NMR spectroscopy provides information about the positions of ^{13}C atoms in a molecule
15.19	be able to use data from ^{13}C NMR spectroscopy to: - i predict the different environments for carbon atoms present in a molecule, given values of chemical shift, δ - ii justify the number of peaks present in a ^{13}C NMR spectrum in terms of the number of carbon atoms in different environments
15.20	be able to use both low and high resolution proton NMR spectroscopy to: - i predict the different types of proton present in a molecule, given values of chemical shift, δ - ii relate relative peak areas, or ratio number of protons, to the relative numbers of ^1H atoms in different environments - iii deduce the splitting patterns of adjacent, non-equivalent protons using the $(n+1)$ rule and hence suggest the possible structures for a molecule - iv predict the chemical shifts and splitting patterns of the ^1H atoms in a given molecule
15.21	know that chromatography separates components of a mixture using a mobile phase and a stationary phase
15.22	be able to calculate R_f values from one-way chromatograms in paper and thin-layer chromatography (TLC) and understand reasons for differences in R_f values
15.23	know that high-performance liquid chromatography, HPLC, and gas chromatography, GC, are types of column chromatography that separate substances because of different retention times in the column and may be used in conjunction with mass spectrometry, in applications such as forensics or drug testing in sport

Unit 5: Transition Metals and Organic Nitrogen Chemistry

IA2 compulsory unit

Externally assessed

Unit description

Introduction

In this unit, the study of electrode potentials builds on the study of redox in Unit 2, including the concept of oxidation number and the use of redox half equations. Students will study further chemistry related to redox, including transition metals.

The organic chemistry section of this unit focuses on arenes and organic nitrogen compounds such as amines, amides, amino acids and proteins. The organic synthesis section requires students to use the knowledge and understanding of organic chemistry that they have gained over the entire specification.

This unit draws on all the other units in the International Advanced Level in Chemistry and students are expected to use their prior knowledge when learning about the areas in this unit. Students will, again encounter ideas of isomerism, bond polarity and bond enthalpy, reagents and reaction conditions, reaction types and mechanisms. Students are also expected to use formulae and balanced equations, and calculate chemical quantities.

Chemistry in action

The study of chemical cells illustrates the impact on scientific thinking when it emerges that ideas developed in different contexts can be shown to be related to a major explanatory principle. In this unit, cell emfs and equilibrium constants are shown to be related to the fundamental criterion for the feasibility of a chemical reaction: the total entropy change.

The explanatory power of the energy-level model for electronic structures is further illustrated by showing how it can help to account for the existence and properties of transition metals. In this context there are opportunities to show the limitations of the models used at this level and to indicate the need for more sophisticated explanations.

Study of the structure of benzene is another example that shows how scientific models develop in response to new evidence. This links to further investigations of the models that chemists use to describe the mechanisms of organic reactions.

The study of catalysts touches on a 'frontier' area for current chemical research and development, which is of theoretical and practical importance. This provides an opportunity to show how the scientific community reports and validates new knowledge.

Practical skills

As in previous units, students can begin their practical work in this unit with some simple test-tube reactions, investigating the reactions of transition metal ions in solution. This may lead to an exploration of redox reactions and, therefore, to the core practical on electrochemical cells.

Skills in volumetric analysis can be consolidated through titrations for redox systems such as iodine-thiosulfate or manganate(VII) titrations.

An opportunity to explore preparative inorganic chemistry is provided in the core practical devoted to making a transition metal complex.

In organic chemistry, there are further functional groups to explore and the possibility of preparing an azo dye.

The final core practical is an organic synthesis and can be used to showcase a selection of the techniques that students have developed to carry out reactions and purify products efficiently and safely.

Mathematical skills

There are opportunities for the development of mathematical skills in this unit. This includes calculating redox potentials, balancing redox equations from half cells, calculating masses and concentrations from redox titrations, investigating the geometry of transition metal complexes, calculating the resonance stability of benzene from thermodynamic data and calculating percentage yields. (Please see *Appendix 6: Mathematical skills and exemplifications* for more detail.)

Assessment information

- First assessment: June 2020.
 - The assessment is 1 hour and 45 minutes.
 - The assessment is out of 90 marks.
 - Students must answer all questions.
 - This paper has three sections:
 - Section A: multiple choice questions
 - Section B: mixture of short-open, open-response, calculations and extended-writing questions
 - Section C: contemporary context question.
 - This paper will contain questions that require information from the Data Booklet (see *Appendix 9*).
 - This paper will include a minimum of 18 marks that target mathematics at Level 2 or above.
 - Students will be expected to apply their knowledge and understanding of experimental methods in familiar and unfamiliar contexts.
 - This paper may contain some synoptic questions which require knowledge and understanding from Units 1, 2 and 4.
 - Calculators may be used in the examination (see *Appendix 8: Use of calculators*).
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